

Bayesian methods and Kernel Ridge Regression applied to global life expectancy

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Machine Learning

1 Dataset Selection and Motivation

For this project, the WHO Life Expectancy dataset (from Kaggle) was selected. The dataset was intentionally chosen due to its noisy and ill-conditioned nature, making it an appropriate benchmark for evaluating Bayesian and kernel-based regression methods in realistic settings. Since life expectancy depends on healthcare, economic, and social factors, the dataset naturally contains complex relationships, redundancy, and non-linear trends.

Exploratory analysis revealed several data quality issues, including inconsistent measurements, redundant variables, and potential data leakage. Also, we identified two inherent structural challenges of the data which are of great interest for our study. In particular, strong collinearity exists between predictors (see figure A.1 in appendix) such as *infant deaths* and *under-five deaths*, which destabilizes Ordinary Least Squares (OLS) regression. Bayesian Ridge Regression addresses this issue through prior-driven regularization, shrinking redundant coefficients and improving stability.

Additionally, many socioeconomic variables exhibit non-linear relationships with life expectancy (see figure A.2 in appendix), motivating the use of Kernel Ridge Regression with an RBF kernel to capture localized non-linear behavior beyond the capabilities of linear models.

Preprocessing of the data A preprocessing pipeline was applied before training. Development status was binarized, country identifiers were removed to avoid geographic memorization, and all variables were standardized using Z-score normalization. This ensured consistent regularization in Bayesian Ridge Regression and prevented large-scale variables from dominating the RBF kernel distance computations.

Overall, the dataset provides a challenging environment for testing the robustness of regularized Bayesian and kernel-based learning methods under realistic real-world conditions.

2 Bayesian Regularization

As seen in Section 1 there are data arrays that are redundant (and, therefore, very collinear) such as the pair 'under-five deaths' and 'infant deaths' or the 'thinness 1-19 years' and 'thinness 1-5 years' one. Taking into account the names and the correlation between pairs, we see that the most conflicting pair in this regard is the one regarding child mortality, since it has a correlation of 1.0.

Once having identified the 'most collinear' pair in the data, and knowing that standard OLS tends to 'panic' and assign heavy opposing weights to highly collinear data, in this section we aim to conduct two experiments to see how the presence of highly collinear data affects the weights of standard OLS and how that effect is mitigated (or fixed; depending on the chosen prior precision) when using Bayesian Ridge Regression to calculate the weights.

2.1 Experiment definitions and chosen methodology

2.1.1 Experiment 1

In this first experiment, we evaluate the impact of prior precision on the posterior weights in the presence of severe multicollinearity. To accomplish this, we compare the coefficient estimates generated by two distinct modeling approaches:

- A model with a very small prior precision ($\lambda \rightarrow 0$), which represents a "flat" or uninformative prior. Mathematically, this is equivalent to a standard Ordinary Least Squares (OLS) regression, as the lack of regularization allows the model to map perfectly to the training data without penalizing weight size.
- A model with a high prior precision ($\lambda \gg 1$), which represents a highly informative, zero-mean Gaussian prior. This acts as a strict Bayesian Ridge Regression, forcing the model to heavily penalize large weights and shrink them toward zero.

To definitively prove that any observed instability in the OLS weights is caused by the multicollinearity trap, we train both models twice: first using the complete dataset (which includes the highly correlated variables), and second after removing the redundant 'under-five deaths' feature. By comparing the results, we expect to observe how OLS panics when presented with redundant data, and how Bayesian regularization successfully compensates for this collinearity.

2.1.2 Experiment 2

The second experiment aims to visualize the "shrinkage path" of the coefficients as the prior precision increases. In *scikit-learn*, the `BayesianRidge` algorithm utilizes an Expectation-Maximization solver that iteratively updates

the prior and noise precisions based on the data. Therefore, this solver prevents strict manual control over the prior precision, even if the iterations are constrained.

To be able to isolate the pure theoretical impact of the prior, we utilize the standard **Ridge** regression algorithm. The Maximum A Posteriori estimate of a Bayesian Linear Regression with an isotropic Gaussian prior is mathematically equivalent to optimizing the standard Ridge cost function:

$$L(w) = ||y - Xw||^2 + \lambda_r ||w||^2, \quad (1)$$

where the standard Ridge penalty parameter represents the ratio of Bayesian precisions ($\lambda_r = \frac{\lambda_p}{\alpha_n}$). In the theoretical framework of Bayesian Linear Regression, the true observation noise precision (α_{noise}) is a property of the data-generating process and acts as a static constant for any given, fixed dataset. Therefore, any increase in the Ridge penalty (λ_r) directly corresponds to a proportional increase in the prior precision (λ_p).

By sweeping through an array of penalty values using the deterministic **Ridge** function, we can rigidly simulate an increasing prior confidence (which is equivalent to a shrinking prior variance, τ^2) and accurately map how the collinear weights collapse as the regularization strength increases.

2.2 Results of Experiment 1 and 2

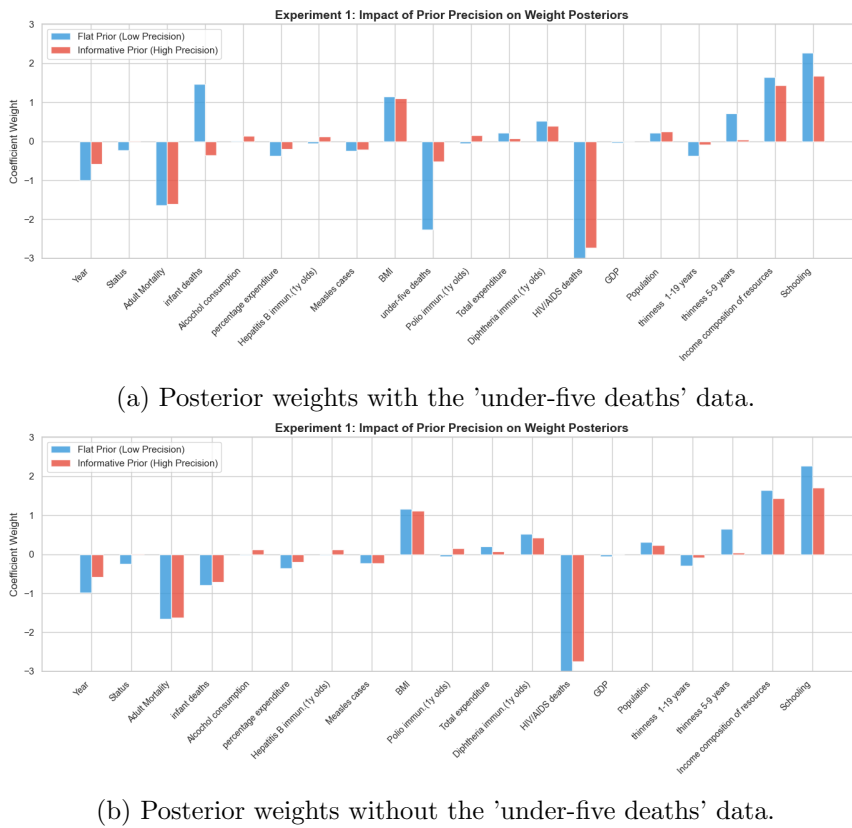
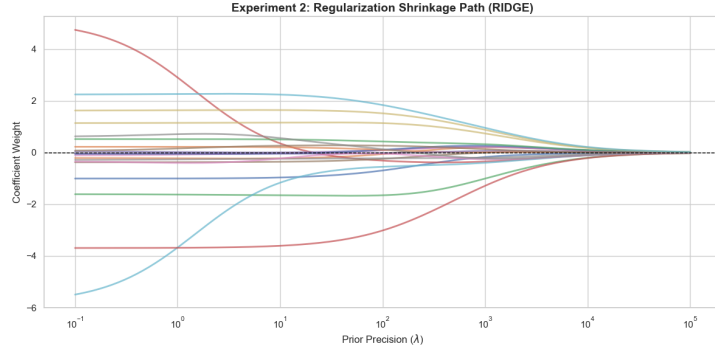
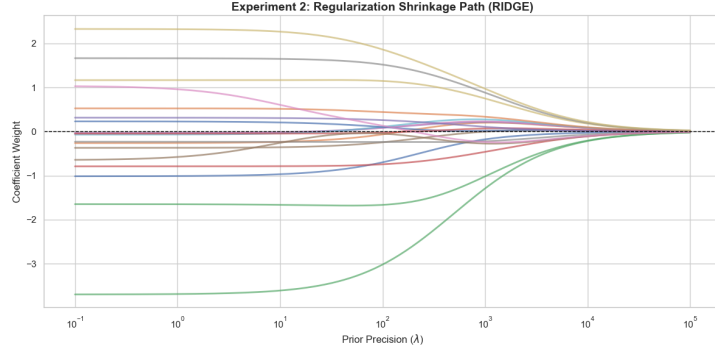


Figure 1: Posterior weights with and without 'under-five deaths' data (Experiment 1).

Figure 1 illustrates the coefficient weights estimated by both a flat prior (Standard OLS) and a highly informative prior (Bayesian Ridge): in Figure 1(a), we observe the models' behavior on the complete dataset containing severe multicollinearity and how, confronted with the redundant variables 'infant deaths' and 'under-five deaths', the OLS model clearly “panics” and assigns them massive, opposing weights. Interpreting this result literally, the OLS coefficients lead to a paradoxical and illogical conclusion: “More infant deaths increase life expectancy, as long as under-five deaths decrease.” On the other hand, the Bayesian Ridge model effectively avoids this problem; the strong prior precision suppresses the noise and shrinks both weights to sensible, stable values.

Figure 1(b) confirms our hypothesis: by deleting the 'under-five deaths' data, we remove the severe collinearity trap and, therefore, the OLS model immediately stabilizes, recognizes that infant deaths negatively impact life expectancy and behaves almost identically to the Bayesian Ridge model. This demonstrates that the initial erratic behavior of the OLS model was indeed caused by multicollinearity: a vulnerability that Bayesian regularization successfully fixes.

Figure 2 maps the trajectory of the coefficient weights as the prior precision (λ) increases: as expected, raising λ forces a shrinkage effect across the coefficients, smoothly driving them toward zero. This occurs because λ acts as a strict mathematical punishment on the L_2 norm of the weight vector ($||w||^2$; see (1)). A comparison between

(a) Posterior weights respect to λ with the 'under-five deaths' data.(b) Posterior weights respect to λ without the 'under-five deaths' data.Figure 2: Posterior weights respect to λ with and without 'under-five deaths' data (Experiment 2).

the two paths reveals the targeted nature of this regularization: in Figure 2(a), where the collinear data pair is present, the artificially inflated weights incur an enormous initial penalty and, as a result, we see the model quickly compensate by aggressively reducing the weights of the troubling pair even at relatively low values of λ . In contrast, Figure 2(b) shows that when the dataset is free of this severe multicollinearity, the naturally stable weights are allowed to persist longer before the increasing λ forces them to shrink.

3 Capturing Non-Linearities

In this section, we apply kernel ridge regression (KRR) to our dataset to study possible nonlinear relationships between the response and explanatory variables. We fit the data within the KRR framework using an optimal choice of hyperparameters and compare its regression performance with that of the optimal Bayesian linear ridge regression developed in the previous section. Finally, we quantify and compare the uncertainty associated with both methods.

3.1 Kernel Ridge Regression

The kernel ridge regression problem for a radial basis function kernel K with length-scale parameter γ and penalization parameter α is equivalent to the finite-dimensional optimization problem

$$\min_{\mathbf{c} \in \mathbb{R}^N} \left\{ \|\mathbf{y} - K_{XX}\mathbf{c}\|^2 + \alpha \mathbf{c}^\top K_{XX}\mathbf{c} \right\}. \quad (2)$$

Note that the regularization parameter α coincides with the residual variance σ^2 of the Gaussian observation noise, and the length-scale parameter γ controls the scale over which variations in the input are expected to be relevant.

This problem admits the closed-form solution

$$\mathbf{c} = (K_{XX}(\gamma) + \alpha I)^{-1} \mathbf{y}, \quad (3)$$

which depends on the hyperparameters α and γ . Details of this derivation are provided in the Appendix.

Optimal hyperparameters α and γ can then be obtained by minimizing the root mean square error (RMSE) on a test data subset. This can be achieved via a grid search over the hyperparameter pair (γ, α) , as illustrated in Figure 3.

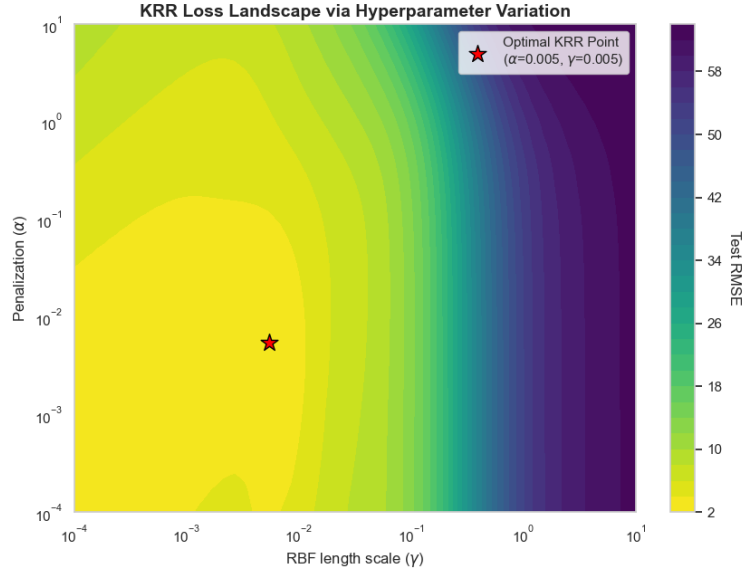


Figure 3: Grid search over the hyperparameter pair (γ, α) for test RMSE minimization.

The KRR with optimal α and γ achieves a test RMSE of 2.4371, which is significantly lower than the RMSE obtained by Bayesian linear ridge regression, which attains a value of 3.6708 for the optimal parameter λ_r . This suggests that KRR captures nonlinear relationships in the data that are not accounted for by the linear regression model.

3.2 Uncertainty Quantification

Additionally, we compare the predictive variance associated with each data point in both Bayesian linear ridge regression and kernel ridge regression. The Bayesian framework naturally provides uncertainty quantification in the linear case through the posterior distribution. Since the KRR solution is mathematically equivalent to the posterior predictive mean of a Gaussian process regression, a corresponding predictive variance can also be derived in a principled manner. Details of the variance computation are provided in the Appendix.

We include a plot comparison of the predictive accuracy and the standard errors of each data point prediction for both methods in Figure 4.

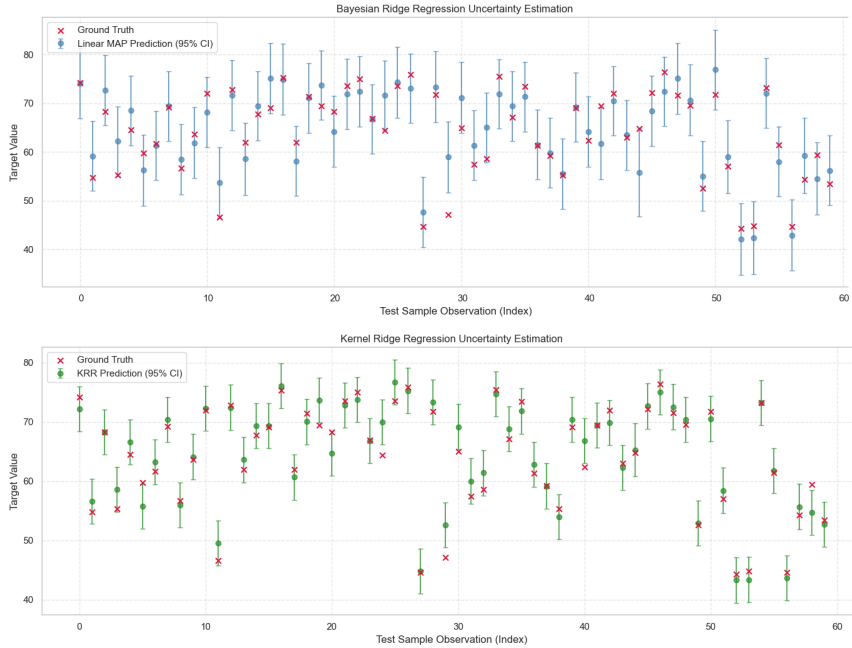


Figure 4: Predictive accuracy and standard errors for the first 60 test data points.

We observe that KRR produces, on average, more accurate predictions than Bayesian linear ridge regression, in accordance with its lower test RMSE. Additionally, it is more confident in its predictions, since the associated standard errors are lower in the KRR case, with an average standard error (over the full dataset) of 1.9472 compared with 3.7019 for Bayesian linear ridge regression.

A Correlation matrix and non-linear dynamics for life expectancy w.r.t GDP per capita

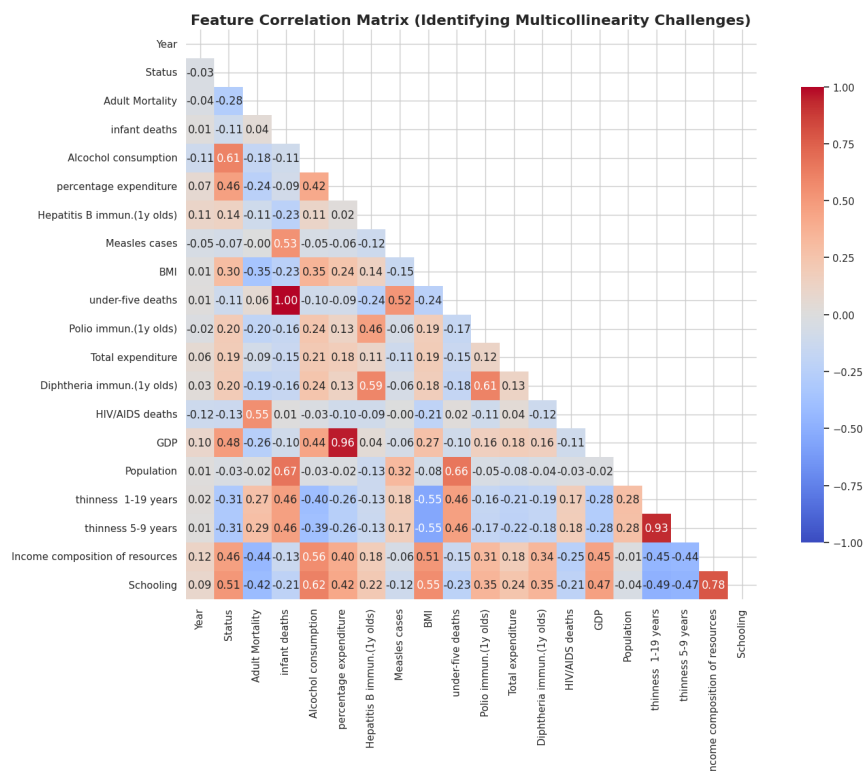


Figure A.1: Correlation matrix for the WHO life expectancy (kaggle) dataset

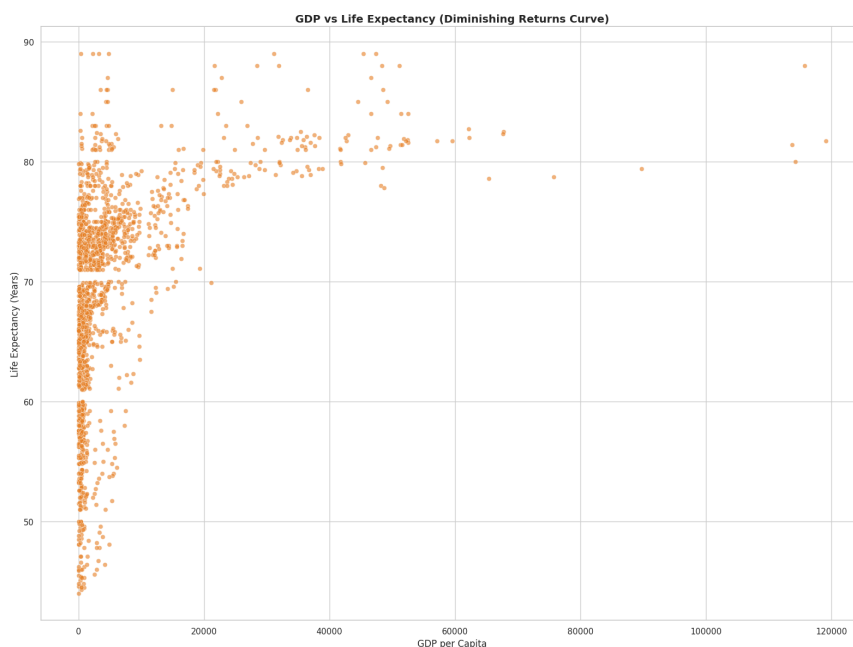


Figure A.2: Diminishing returns for wealth: life expectancy with respect to GDP per capita. A sharp increase in life expectancy, which quickly plateoes, occurs as GDP increases. This is a well-known example of non-linear dynamics of some macro-economics variables

B Kernel Ridge Regression Solution

We recall that the regularized empirical risk minimization problem for a general non-linear regression model with Gaussian observation noise can be stated as

$$\min_{f \in \mathcal{H}} \frac{1}{2\sigma^2} \sum_{i=1}^N (y_i - f(x_i))^2 + \frac{1}{2} \|f\|_{\mathcal{H}}^2 = \min_{f \in \mathcal{H}} \frac{1}{2} \sum_{i=1}^N (y_i - f(x_i))^2 + \frac{\alpha}{2} \|f\|_{\mathcal{H}}^2, \quad (4)$$

where $(\mathcal{H}, \|\cdot\|_{\mathcal{H}})$ is a normed function space, σ^2 is the residual variance (i.e. the variance of the Gaussian observation noise), and $\alpha > 0$ is the penalization parameter, which, in this formulation, coincides with the residual variance ($\alpha = \sigma^2$).

Let us assume that $\mathcal{H}$ is a reproducing kernel Hilbert space with a radial basis function (RBF) kernel,

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2), \quad (5)$$

where $\gamma > 0$ is the length-scale parameter. Then, by the representer theorem, the solution of the optimization problem admits the finite-dimensional representation

$$f(x) = \sum_{j=1}^N c_j K(x_j, x) = K_{Xx}^T \mathbf{c}, \quad (6)$$

where $\mathbf{c} = (c_1, \dots, c_N)^T$ is given by

$$\mathbf{c} = (K_{XX}(\gamma) + \alpha I)^{-1} \mathbf{y}, \quad (7)$$

which is the solution of the equivalent finite-dimensional optimization problem

$$\min_{\mathbf{c} \in \mathbb{R}^N} \left\{ \|\mathbf{y} - K_{XX} \mathbf{c}\|^2 + \alpha \mathbf{c}^T K_{XX} \mathbf{c} \right\}. \quad (8)$$

Therefore, for fixed hyperparameters, the optimization problem admits an analytical solution, and predictions are obtained according to expression (6).

C Variance Computation

In the Bayesian linear ridge regression framework, an isotropic Gaussian prior with variance τ^2 is assumed for the weights. Under Gaussian observation noise, this leads to a Gaussian posterior. In this setting, the posterior mean coincides with the solution of a linear ridge regression problem with regularization parameter $\lambda_r = \frac{\sigma^2}{\tau^2}$, and the posterior covariance matrix is given by

$$\Sigma_{\text{Linear}} = \left(\frac{1}{\sigma^2} X^T X + \frac{1}{\tau^2} I \right)^{-1}. \quad (9)$$

The variance associated with the prediction y_* at an individual test point x_* is then given by

$$\sigma_{\text{Linear}}^2(x_*) = x_*^T \Sigma_{\text{Linear}} x_* + \sigma^2. \quad (10)$$

For the kernel ridge regression case, we draw an equivalence with the Gaussian process regression Bayesian framework. When a Gaussian process is assumed as a prior for the regression function f in (4) with zero prior mean function, then, for a general nonlinear regression model with Gaussian observation noise, the posterior is also a Gaussian process. The posterior mean coincides with the KRR solution with regularization parameter $\alpha = \sigma^2$, and the posterior covariance function is given by

$$\Sigma_{\text{Kernel}}(x_*, x'_*) = K(x_*, x'_*) - \mathbf{k}_*^T (K_{XX} + \sigma^2 I)^{-1} \mathbf{k}_*. \quad (11)$$

The variance associated with the prediction y_* at an individual test point x_* is then given by

$$\sigma_{\text{Kernel}}^2(x_*) = \Sigma_{\text{Kernel}}(x_*, x_*) + \sigma^2. \quad (12)$$

These variance decompositions follow from the fact that the observation model can be written as $y_* = f(x_*) + \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ is independent of $f(x_*)$. Consequently, we obtain $\text{Var}(y_*) = \text{Var}(f(x_*)) + \sigma^2$, where $\text{Var}(f(x_*))$ corresponds to the posterior variance at the point x_* .